

Lesson 7: Introduction to Confidence Intervals

BSTA 551: Statistical Inference

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Lesson 7: From Point Estimates to Intervals

Review: What We've Learned About Estimation

Point Estimation (Lessons 1-6):

- Estimators: statistics that estimate parameters
- Properties: bias, variance, MSE, efficiency
- Construction: Method of Moments, Maximum Likelihood
- Quality: MVUE, sufficient statistics, Cramér-Rao bound

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The Limitation of Point Estimates:

A single number ($\hat{\theta} = 0.75$) tells us nothing about **precision** or **uncertainty** of the estimate.

Today's Goals



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4. Construct and interpret z-intervals (known σ)
5. Determine sample sizes for desired precision
6. Understand one-sided confidence bounds

Motivating Example: Clinical Trial

A new antihypertensive drug is tested in $n = 36$ patients. The mean reduction in systolic blood pressure is $\bar{x} = 12.3$ mmHg.

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Questions we can't answer with just the point estimate:

- How close is 12.3 to the true mean reduction?
- Is the true mean likely to be at least 10 mmHg?
- How much could $\bar{X}$ vary in repeated trials?

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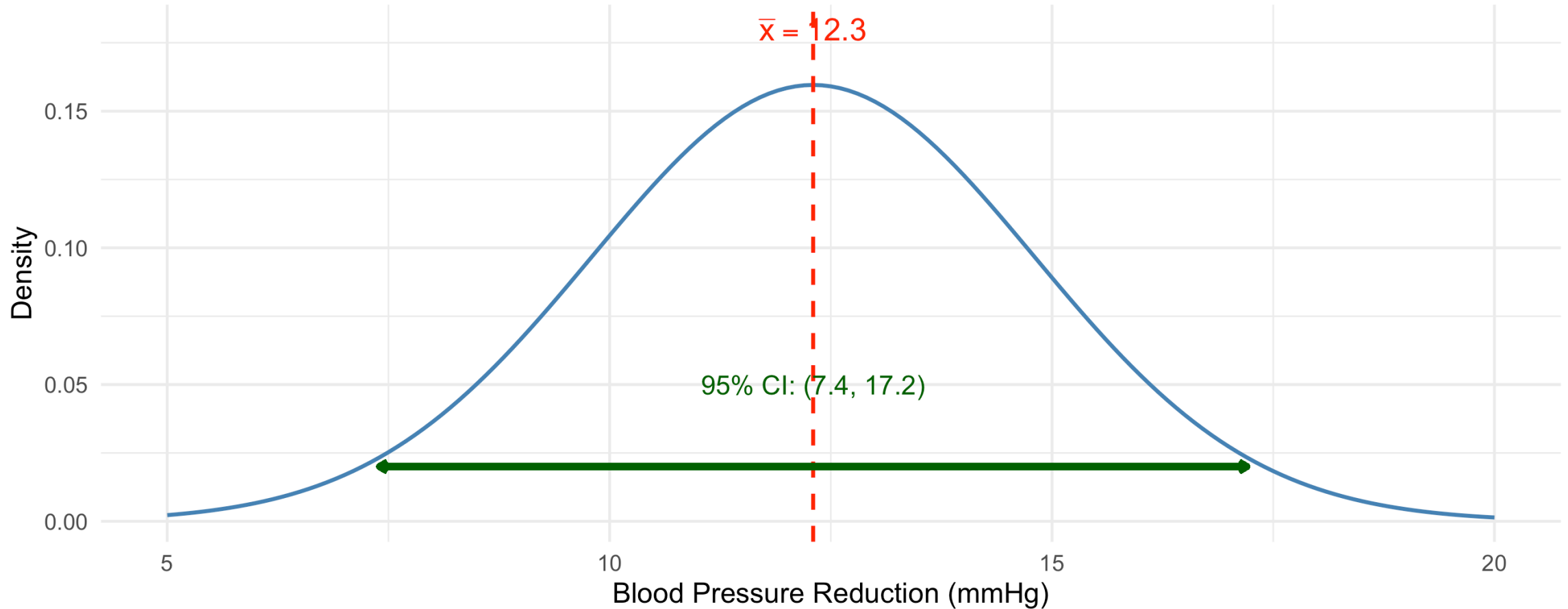
The Solution

A **confidence interval** provides a range of plausible values for the parameter, along with a measure of confidence.

The Fundamental Idea

From Point to Interval

Point Estimate with Confidence Interval



Key insight: The interval captures our uncertainty about the true parameter value.

What Is a Confidence Interval?

! Definition: Confidence Interval

A **confidence interval (CI)** for a parameter θ is a random interval (L, U) computed from sample data such that:

$$P(L < \theta < U) = 1 - \alpha$$

for a specified confidence level $1 - \alpha$.

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Important:

- L and U are **random** (they depend on the sample)
- θ is **fixed** (but unknown)
- The probability statement is about the **random event** that the lower and upper bound cover θ , not about any randomness in θ itself.
- The probability statement only makes sense when thinking about defining the estimators L and U and their sampling distributions, it does not make sense when talking about one sample and the observed values for L and U

Correct Interpretation of Confidence Intervals

Common Misconception

“There is a 95% probability that μ lies in the interval (7.4, 17.2).”

This is WRONG! After computing the interval, μ either is or is not in the interval. There's no probability involved.

Correct Interpretation of Confidence Intervals

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Correct Interpretation

“If we repeated this procedure many times, 95% of the resulting intervals would contain the true μ .”

Or more simply/loosely: “We are 95% confident that the true mean lies between 7.4 and 17.2.”

Common Mistakes to Avoid (Chihara Example 7.2)

Suppose a 95% CI for mean gas mileage is (29.5, 33.4) mpg. Evaluate:

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 - **Wrong:** CI is for population *mean*, not individual values.
2. ❌ “95% of all samples will give an average mileage between 29.5 and 33.4 mpg.”
 - **Wrong:** Each sample gives a *different* CI; 95% of *intervals* contain μ .

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2. ❌ “95% of all samples will give an average mileage between 29.5 and 33.4 mpg.”
 - **Wrong:** Each sample gives a *different* CI; 95% of *intervals* contain μ .
3. ❌ “There is a 95% chance that the true mean is between 29.5 and 33.4 mpg.”
 - **Wrong:** μ is fixed, not random. It’s either in the interval or not.

Visualizing Confidence Level: Simulation

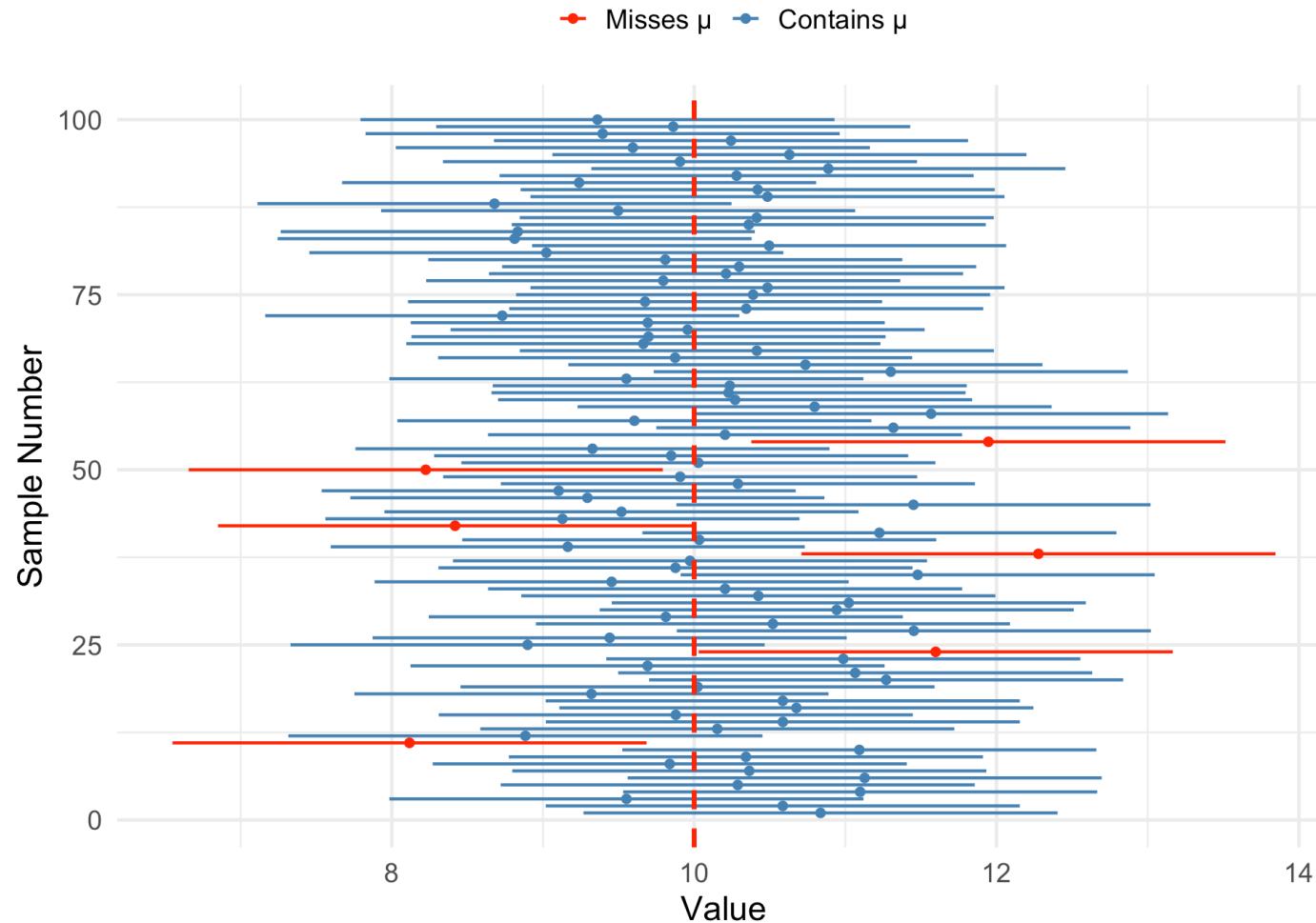
```
1 # Simulate 100 confidence intervals
2 set.seed(551)
3 true_mu <- 10; sigma <- 4; n <- 25
4
5 ci_data <- tibble(sample_id = 1:100) |>
6   rowwise() |>
7   mutate(
8     xbar = mean(rnorm(n, true_mu, sigma)),
9     lower = xbar - 1.96 * sigma / sqrt(n),
10    upper = xbar + 1.96 * sigma / sqrt(n),
11    contains_mu = lower < true_mu & upper > true_mu
12  )
13
14 # Count how many contain the true mean
15 sum(ci_data$contains_mu)
```

```
[1] 94
```

Visualizing Confidence Level: The Plot

50 Confidence Intervals (95% Level)

True $\mu = 10$ (red dashed line)



The R Code for CI Visualization

```
1 # Simulate *5000* confidence intervals
2 set.seed(551)
3 true_mu <- 10; sigma <- 4; n <- 25
4
5 ci_data <- tibble(sample_id = 1:5000) |>
6   rowwise() |>
7   mutate(
8     xbar = mean(rnorm(n, true_mu, sigma)),
9     lower = xbar - 1.96 * sigma / sqrt(n),
10    upper = xbar + 1.96 * sigma / sqrt(n),
11    contains_mu = lower < true_mu & upper > true_mu
12  )
13
14 # Proportion containing true mu (should be ~95%)
15 mean(ci_data$contains_mu)
```

```
[1] 0.9498
```

This is how the confidence level works: about 95% of intervals capture μ .

Your Turn: CI Interpretation

Exercise: A researcher reports a 95% CI for mean hospital stay after surgery as (3.2, 4.8) days. Which statements are correct?

- A. "95% of patients stay between 3.2 and 4.8 days"
- B. "We are 95% confident the true mean stay is between 3.2 and 4.8 days"
- C. "There is a 95% probability the mean is in this interval"
- D. "If we repeated this study many times, about 95% of the CIs would contain the true mean"

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- C. "There is a 95% probability the mean is in this interval"
- D. "If we repeated this study many times, about 95% of the CIs would contain the true mean"

Answer: B and D are correct interpretations.

Pivotal Quantities

The Key to Constructing CIs

! Definition: Pivotal Quantity

A **pivotal quantity** is a function $h(X_1, \dots, X_n; \theta)$ of the data and parameter such that:

1. It depends on both the sample and θ
2. Its **distribution does not depend on θ** (or any other unknown parameters)

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Why pivotal quantities matter:

- We can find critical values that don't depend on unknowns
- We can “solve for θ ” to get confidence limits

Example: The Standard Pivotal Quantity

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Z is a pivotal quantity!

From Pivotal Quantity to CI

Since $Z \sim N(0, 1)$, we know:

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Solve for μ :

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

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What is L and U here?

The General Pivotal Quantity Method

Recipe for Constructing CIs

1. Find a pivotal quantity $h(X_1, \dots, X_n; \theta)$
2. Find constants a and b such that $P(a < h < b) = 1 - \alpha$
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Example (Devore 8.1): For normal data with known σ :

- Pivotal: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
- Constants: $a = -z_{\alpha/2}, b = z_{\alpha/2}$
- Solve: $\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

CI for Mean: σ Known

The Z-Interval

! Z Confidence Interval for Normal Mean (σ Known)

If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ known, a $100(1 - \alpha)\%$ CI for μ is:

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the critical value with $P(Z > z_{\alpha/2}) = \alpha/2$.

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where $z_{\alpha/2}$ is the critical value with $P(Z > z_{\alpha/2}) = \alpha/2$.

Common critical values:

Confidence Level	α	$\alpha/2$	$z_{\alpha/2}$
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576



Example: Tensile Strength (Devore Example 8.1)

Synthetic fibers are tested for tensile strength. From past data, $\sigma = 55$ MPa is known. A sample of $n = 31$ fibers has $\bar{x} = 1064$ MPa.

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95% Confidence Interval:

```
1 xbar <- 1064
2 sigma <- 55
3 n <- 31
4 alpha <- 0.05
5
6 z_crit <- qnorm(1 - alpha/2)
7 margin <- z_crit * sigma / sqrt(n)
8
9 c(xbar - margin, xbar + margin)

[1] 1044.639 1083.361
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Interpretation: We are 95% confident that the true mean tensile strength is between 1044.6 and 1083.4 MPa.

Medical Example: Blood Glucose Levels

A hospital monitors fasting blood glucose in diabetic patients. Historical data shows $\sigma = 18$ mg/dL. A sample of $n = 40$ patients has $\bar{x} = 142$ mg/dL.

Construct a 99% CI for the population mean glucose level:

```
1 xbar <- 142
2 sigma <- 18
3 n <- 40
4 alpha <- 0.01
5
6 z_crit <- qnorm(1 - alpha/2)
7 cat("z critical value:", round(z_crit, 3), "\n")
```

z critical value: 2.576

```
1 margin <- z_crit * sigma / sqrt(n)
2 cat("Margin of error:", round(margin, 2), "mg/dL\n")
```

Margin of error: 7.33 mg/dL

```
1 cat("99% CI: (", round(xbar - margin, 1), ",", round(xbar + margin, 1), ")")
```

99% CI: (134.7 , 149.3)

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Margin of error: 7.33 mg/dL

```
1 cat("99% CI: (", round(xbar - margin, 1), ",", round(xbar + margin, 1), ")")
```

99% CI: (134.7 , 149.3)

Clinical interpretation: We can be 99% confident that the true mean fasting glucose for this patient population is between 134.7 and 149.3 mg/dL.

Anatomy of a Confidence Interval

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Point Estimate Margin of Error

$\bar{x}$

$$z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Three factors affect the margin of error:

1. **Confidence level** ($z_{\alpha/2}$): Higher confidence → wider interval → less precise
2. **Population variability** (σ): More variation → wider interval → less precise
3. **Sample size** (n): Larger sample → narrower interval → more precise

Interactive: Effect of Confidence Level

```
1 # Same data, different confidence levels
2 xbar <- 1064; sigma <- 55; n <- 31
3
4 confidence_levels <- c(0.90, 0.95, 0.99)
5
6 tibble(conf_level = confidence_levels) |>
7   mutate(
8     z_crit = qnorm(1 - (1 - conf_level)/2),
9     margin = z_crit * sigma / sqrt(n),
10    lower = xbar - margin,
11    upper = xbar + margin,
12    width = upper - lower
13  )
```

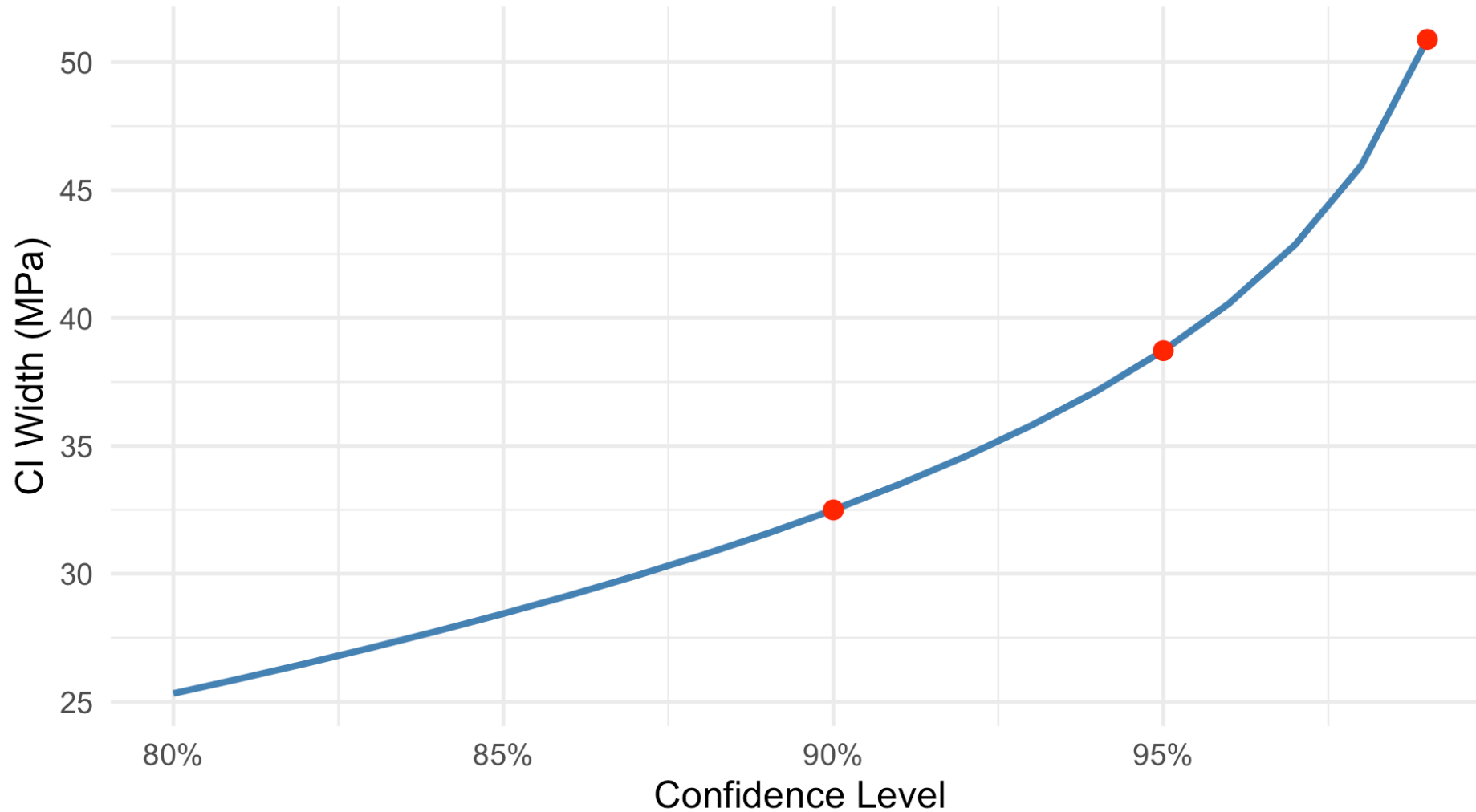
A tibble: 3 × 6

	conf_level	z_crit	margin	lower	upper	width
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.9	1.64	16.2	1048.	1080.	32.5
2	0.95	1.96	19.4	1045.	1083.	38.7
3	0.99	2.58	25.4	1039.	1089.	50.9

Takeaway: Higher confidence requires wider intervals!

Visualizing the Trade-off

Trade-off: Confidence Level vs. Precision



Sample Size Determination

Question: How large should n be to achieve a desired precision?

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If we want the margin of error to be at most b :

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Solving for n :

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{b} \right)^2$$

Sample Size Example: Clinical Trial Design

A pharmaceutical company wants to estimate mean drug clearance time. From pilot data, $\sigma \approx 25$ minutes. They want a 95% CI with margin of error ≤ 5 minutes.

```
1 sigma <- 25
2 b <- 5 # desired margin of error
3 alpha <- 0.05
4 z_crit <- qnorm(1 - alpha/2)
5
6 n_required <- (z_crit * sigma / b)^2
7 cat("Minimum sample size:", ceiling(n_required))
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Minimum sample size: 97

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7 cat("Minimum sample size:", ceiling(n_required))
```

Minimum sample size: 97

Note: Always round **up** to ensure the desired precision!

Sample Size: Effect of Requirements

```
1 # How does required n change with margin of error and confidence?
2 expand_grid(
3   margin = c(3, 5, 10),
4   conf_level = c(0.90, 0.95, 0.99)
5 ) |>
6 mutate(
7   z = qnorm(1 - (1 - conf_level)/2),
8   n_required = ceiling((z * 25 / margin)^2)
9 ) |>
10 pivot_wider(names_from = margin, values_from = n_required,
11             names_prefix = "margin_") |>
12 select(-z)
```

A tibble: 3 × 4

	conf_level	margin_3	margin_5	margin_10
	<dbl>	<dbl>	<dbl>	<dbl>
1	0.9	188	68	17
2	0.95	267	97	25
3	0.99	461	166	42

Key insight: Halving the margin of error (from 10 to 5) quadruples the sample size!

One-Sided Confidence Bounds

Sometimes we only care about one direction:

- **Lower bound:** We want to know the minimum plausible value
- **Upper bound:** We want to know the maximum plausible value

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One-Sided Bounds

Lower Confidence Bound: $\bar{x} - z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} < \mu$

Upper Confidence Bound: $\mu < \bar{x} + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$

Note: Use z_{α} (not $z_{\alpha/2}$) for one-sided bounds at confidence level $1 - \alpha$.

Why z_α instead of $z_{\alpha/2}$?

For a **two-sided** CI, we split α into two tails ($\alpha/2$ each):

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

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For a **one-sided lower bound**, all α goes in one tail:

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_\alpha\right) = 1 - \alpha$$

Rearranging for μ :

$$P\left(\bar{X} - z_\alpha \cdot \frac{\sigma}{\sqrt{n}} < \mu\right) = 1 - \alpha$$

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Rearranging for μ :

$$P\left(\bar{X} - z_\alpha \cdot \frac{\sigma}{\sqrt{n}} < \mu\right) = 1 - \alpha$$

So the lower bound is: $L = \bar{X} - z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

Key insight: One-sided bounds use z_α because we put all the “error probability” α on one side, not split between two sides.



Medical Example: Drug Half-Life

A pharmacologist measures half-life of a new drug. With $n = 20$ subjects, $\bar{x} = 4.2$ hours, and known $\sigma = 0.8$ hours.

Find a 95% lower confidence bound:

```
1 xbar <- 4.2
2 sigma <- 0.8
3 n <- 20
4 alpha <- 0.05
5
6 z_crit <- qnorm(1 - alpha) # Note: z_alpha, not z_alpha/2
7 lower_bound <- xbar - z_crit * sigma / sqrt(n)
8
9 cat("95% lower confidence bound:", round(lower_bound, 2), "hours\n")
```

95% lower confidence bound: 3.91 hours

```
1 cat("Interpretation: We are 95% confident that the true half-life",
2     "\nis at least", round(lower_bound, 2), "hours")
```

Interpretation: We are 95% confident that the true half-life is at least 3.91 hours

Why Lower vs. Upper Bounds?

Use a **LOWER** bound when:

- Testing if a treatment effect is “at least” some value
- Establishing minimum efficacy
- Setting safety margins

Why Lower vs. Upper Bounds?

Use a **LOWER** bound when:

- Testing if a treatment effect is “at least” some value
- Establishing minimum efficacy
- Setting safety margins

Use an **UPPER** bound when:

- Estimating maximum risk or adverse event rate
- Establishing worst-case scenarios
- Quality control (maximum defect rate)

Example: Remission Time (Devore 8.7)

A surgeon wants a lower bound for mean remission time after cancer surgery. From $n = 25$ patients, $\bar{x} = 42$ months, $\sigma = 12$ months (from historical data).

```
1 xbar <- 42
2 sigma <- 12
3 n <- 25
4 alpha <- 0.05
5
6 # 95% lower confidence bound
7 z_alpha <- qnorm(1 - alpha)
8 lower <- xbar - z_alpha * sigma / sqrt(n)
9
10 cat("95% lower confidence bound:", round(lower, 1), "months\n")
```

95% lower confidence bound: 38.1 months

```
1 cat("\nWe are 95% confident that true mean remission time\n")
```

We are 95% confident that true mean remission time

```
1 cat("is at least", round(lower, 1), "months.")
```

is at least 38.1 months.

Pivotal Quantities for Other Distributions

Example: Exponential Distribution (Devore 8.5)

Time-to-breakdown of insulating fluid follows $\text{Exp}(\lambda)$ with mean $1/\lambda$ and variance $1/\lambda^2$. Given $n = 10$ observations with $\sum x_i = 550.87$ min.

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The pivotal quantity:

$$h = 2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$$

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The pivotal quantity:

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Why is this pivotal?

- Depends on data ($\sum X_i$) and parameter (λ)
- Distribution (χ_{2n}^2) doesn't depend on λ

Exponential CI Derivation

Let $\chi^2_{.025,20}$ and $\chi^2_{.975,20}$ be the critical values. Then:

$$P\left(\chi^2_{.975,20} < 2\lambda \sum X_i < \chi^2_{.025,20}\right) = 0.95$$

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Let $\chi^2_{.025,20}$ and $\chi^2_{.975,20}$ be the critical values. Then:

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Solving for λ :

$$P\left(\frac{\chi^2_{.975,20}}{2 \sum X_i} < \lambda < \frac{\chi^2_{.025,20}}{2 \sum X_i}\right) = 0.95$$

```
1 sum_x <- 550.87
2 lower <- qchisq(0.025, df = 20) / (2 * sum_x)
3 upper <- qchisq(0.975, df = 20) / (2 * sum_x)
4 cat("95% CI for λ: (", round(lower, 4), ",", round(upper, 4), ")")
```

95% CI for λ : (0.0087 , 0.031)

CI for Mean of Exponential

Since the mean of $\text{Exp}(\lambda)$ is $\mu = 1/\lambda$:

```
1 # CI for mean breakdown time
2 mean_lower <- 1/upper
3 mean_upper <- 1/lower
4
5 cat("95% CI for mean breakdown time:\n")
```

95% CI for mean breakdown time:

```
1 cat("(", round(mean_lower, 1), ",", round(mean_upper, 1), ") minutes")
```

(32.2 , 114.9) minutes

CI for Mean of Exponential

Since the mean of $\text{Exp}(\lambda)$ is $\mu = 1/\lambda$:

```
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```

95% CI for mean breakdown time:

```
1 cat("(", round(mean_lower, 1), ",", round(mean_upper, 1), ") minutes")
( 32.2 , 114.9 ) minutes
```

Note: The interval is quite wide — reflecting variability in the data and small sample size.

Summary and Looking Ahead

Lesson 7 Summary

Key Concepts:

1. **Confidence Intervals** provide a range of plausible values with quantified uncertainty
2. **Correct Interpretation:** “95% of intervals constructed this way contain θ ”
3. **Pivotal Quantities** allow CI construction without knowing unknown parameters
4. **Z-intervals** (σ known): $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
5. **Sample Size:** $n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{b} \right)^2$
6. **One-sided bounds** use z_{α} instead of $z_{\alpha/2}$

Key Formulas for Z-Intervals

Quantity	Formula
Two-sided CI	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
Lower bound	$\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$
Upper bound	$\bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$
Sample size	$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{b} \right)^2$
Width	$w = 2z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Looking Ahead: Lesson 8

Next Lesson: One-Sample t-Intervals and Proportions

- What if σ is unknown? (The t-interval)
- The Student's t distribution
- Confidence intervals for population proportions
- Wald vs. Score intervals
- Checking assumptions

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 8.1
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 7.1-7.2

Questions?

Thank you!